UNIVERSITY PROJECT DEFENSE / BIG DATA SYSTEMS

NYC Mobility Intelligence

Urban Mobility, Demand Forecasting and Explainable Surge Pricing

NYC Yellow Taxi Big Data Analytics - Apache Spark - SQLite - Streamlit

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Why NYC Mobility Analytics?

Fragmented mobility signals

Millions of trips contain demand, congestion and pricing patterns that manual analysis cannot reveal.

Operational planning gap

Operators need zone and hour level insight to anticipate demand and explain pricing decisions.

Batch and live views

Historical analytics and simulated dispatch events require separate but connected processing paths.

WHY THIS QUALIFIES AS BIG DATA

29.7M
raw trip records

488 MiB
compressed Parquet

Volume: tens of millions of observations
Variety: temporal, spatial, fare and passenger fields
Velocity: simulated event ingestion for live analytics
Processing: distributed transformations and aggregations

Project Objectives

01 Distributed ETL

Validate, clean and aggregate eight monthly Parquet files with PySpark.

02 Demand analytics

Measure hourly demand, congestion, revenue and pickup hotspots.

03 Pricing intelligence

Apply percentile-based surge tiers with reasons and fare impact.

04 Interactive UI

Present historical, forecast and simulated live metrics in Streamlit.

BATCH	**TLC Parquet > Spark ETL > Aggregated CSV > SQLite > Dashboard**
LIVE DEMO	**Trip templates > Event generator > SQLite live buffer > Auto-refresh UI**

Live events are simulated from historical trip templates. No live TLC GPS feed is used.

NYC Yellow Taxi Trip Records

Official source

NYC Taxi and Limousine Commission Trip Record Data
Eight monthly Yellow Taxi Parquet files: January to August 2026

What each trip describes

Pickup and drop-off timestamps
Pickup and drop-off Taxi Zone IDs
Passenger count and trip distance
Fare, tips and total amount
Rate and payment attributes

Taxi Zone lookup: borough, zone name and service zone

Local dataset scale

29,703,355 raw records
511,451,602 bytes, approximately 488 MiB
261 pickup zones represented in the serving layer

Source: NYC TLC Trip Record Data and Yellow Taxi Trip Records Data Dictionary

From Raw Trips to Analytical Tables

01 Schema validation

Require pickup/drop-off timestamps, passenger count, distance, zone IDs, fare and total amount.

Business rules

Passengers 1-6 / Fare $2.50-$500 / Distance 0.10-150 miles / Duration 1-300 minutes / Valid Taxi Zone IDs 1-265

02 Quality filters

Remove null locations and invalid trips. Enforce passenger, fare, distance and duration limits.

Spatial enrichment

Join the official Taxi Zone lookup to attach zone name, borough and service zone to every pickup location.

03 Feature engineering

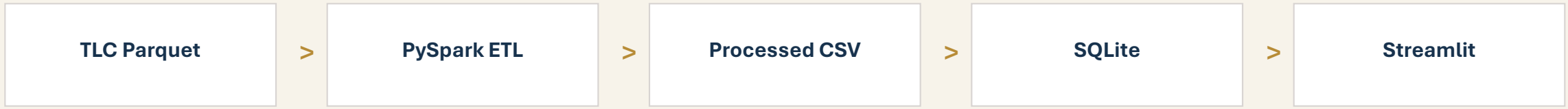
Calculate duration, hour, weekday, date, month, weekend flag and speed in mph.

Aggregation outputs

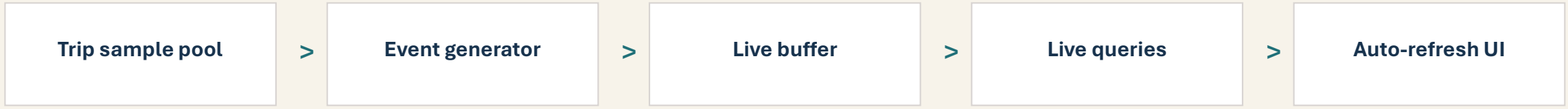
168 weekday-hour rows / 261 zone summaries / 36,947 zone-hour records / 244 daily trend rows / 36,947 pricing windows

Batch Analytics and Simulated Live Operations

HISTORICAL BATCH PATH



SIMULATED LIVE PATH



Serving layer

SQLite stores historical aggregates, pricing windows and the bounded live_trips event buffer.

Separation of concerns

Historical metrics remain distinct from simulated live events to preserve data provenance.

Tools Used Across the Pipeline

Processing

Python 3.12
Apache Spark / PySpark 3.5.1
Pandas, PyArrow and scikit-learn

Storage and serving

Parquet raw data lake
CSV analytical exports
SQLite with SQLAlchemy

Visualization

Streamlit
Plotly interactive charts
Folium geospatial maps

Analytics logic

Gradient-boosting demand model
Seasonal comparison baseline
Median-based spike detection

Quality assurance

Pytest regression tests
Schema and null validation
Database integrity checks

Application design

Parameterized SQL queries
Cached database connection
Auto-refreshing live fragment

Demand Forecasting, Spike Detection and Surge Pricing

Demand forecast

Model: histogram gradient-boosting regressor
Features: hour, weekday, month, weekend and borough
Output: next six hours with an uncertainty band

Trained on January-July data and tested chronologically on August.

Demand-spike detection

Baseline: median trip demand for each pickup zone
Alert: at least 35% above baseline
Severity: Elevated or Critical

Shows observed demand, normal demand and percentage increase.

Explainable surge pricing

Below P50: 1.0x
P50 and above: 1.3x
P75 and above: 1.8x
P90 and above: 2.5x

The dashboard explains the threshold and fare impact for every decision.

MODEL EVALUATION

Chronological holdout: ML WAPE 14.99% versus 18.49% for the seasonal baseline. Pricing and live events remain simulations.

Six Interactive Views in Streamlit

Executive Intelligence

KPIs, six-hour forecast and priority demand-spike alerts.

Demand and Congestion

Weekday-hour heatmap and traffic-speed profile.

Pickup Hotspots

Ranked zones and an interactive NYC map.

Explainable Pricing

Zone, weekday and hour controls with decision rationale.

Daily Trends

Trip volume, gross revenue, fare and distance over time.

Live Operations

Streaming rate, live revenue, hotspots, alerts and recent dispatches.

LIVE DEMONSTRATION

Start or pause the background generator, inject a 50-ride burst, watch the two-second refresh, and explain how burst intensity changes the pricing tier.

Processed Data and Delivered Analytics

29.7M	21.0M	$618.5M	261
RAW TRIP RECORDS	RETAINED TRIPS IN SERVING SUMMARIES	AGGREGATED GROSS REVENUE	PICKUP ZONES

Implemented capabilities

Distributed cleaning and aggregation with PySpark
Zone and hour level demand analysis
Percentile-based explainable pricing
Trained ML six-hour forecast with uncertainty
Demand-spike alerts and simulated live operations

Serving layer evidence

168 weekday-hour demand rows
36,947 zone-hour demand rows
36,947 pricing decision windows
244 daily observations
Top zones: Upper East Side South, JFK Airport and Midtown Center

Metrics read from the local Parquet files and db/mobility.db serving database.

Current Boundaries and Next Steps

Achievements

End-to-end batch pipeline
Interactive six-view dashboard
Explainable percentile pricing
Evaluated gradient-boosting forecast
Automated dashboard and query tests

Limitations

Completed trips do not reveal unmet demand
No driver-supply or customer-response data
Live events are simulated
ML uses calendar and borough features only
SQLite targets local demonstration scale

Future scope

Connect weather and event data
Add driver-supply signals
Compare LSTM and XGBoost models
Use Kafka for external event streams
Deploy PostgreSQL/PostGIS in the cloud

Future-scope items are proposals and are not implemented in the current project.

NYC Mobility Intelligence

29.7 million public taxi trips become decision-ready mobility analytics through a reproducible batch pipeline.

The project demonstrates distributed ETL, spatial and temporal aggregation, a trained demand forecast, explainable surge pricing, spike detection and simulated live operations.

The dashboard connects each analytical output to an operational question while keeping simulated and historical data separate.

THANK YOU

Questions and Answers